Enhancing Logo Recognition and Similarity Search Through Synthetic Data Generation and Deep Learning Technique

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Abstract

This study proposes a scalable and weakly supervised framework for logo recognition and similarity search, designed to address the limitations of conventional datasets in terms of scale, annotation cost, and generalization. The approach begins by generating a large-scale synthetic dataset containing 1.7 million logos, each paired with a descriptive textual prompt. To construct a rich semantic representation, each logo is embedded using a multimodal fusion of visual features from a frozen ResNet50 network and linguistic features from a MiniLM transformer, resulting in a 2432-dimensional vector. Dimensionality reduction via UMAP, followed by hierarchical density-based clustering using HDBSCAN, yields over 209,000 pseudo-categories that serve as weak labels for subsequent metric learning. A Triplet Network is trained using these labels to produce a compact, 256-dimensional embedding space optimized through triplet margin loss. Quantitative evaluation, based on FAISS-powered nearest-neighbor retrieval, demonstrates that the ResNet50-based model outperforms VGG16 and EfficientNet-B0 across Precision@K, Recall@K, and F1@K metrics. Qualitative analysis further reveals strong generalization and structural robustness, particularly in identifying visually coherent logo variants and near-duplicates. The proposed system effectively bridges the gap between high-volume synthetic data and practical retrieval tasks, offering a compelling solution for real-world applications such as brand monitoring, duplicate detection, and visual content discovery. By combining multimodal representation, unsupervised clustering, and deep metric learning, the framework achieves high retrieval performance without requiring human-annotated labels, highlighting the viability of weak supervision for large-scale visual understanding.

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Chapter I

Introduction

The logo recognition and similarity search that highly crucial in computer vision, digital marketing, and finding applications in brand authentication, fake detection, and intellectual property protection. With the increasing use of logos across industries, the maintenance of accuracy and efficient recognition systems is becoming more challenging. Due to their reliance on manually engineered features, the conventional recognition methods are often deficient in robustness to real-world variations like transformations, occlusions, and distortions.

Deep learning has significantly advanced logo recognition, as a result of convolutional neural networks (CNNs) to enhance the learning of hierarchical feature representations, directly from the raw images. ResNet50 has now gained crucial significance among the various deep learning architectures, due to its residual framework of learning, which typically allows the deep networks to train efficiently. However, the limited availability of labeled datasets is a significant barrier towards the process of training deep learning models training, for logo recognition. Provided that there is high diversity and continuous emergence of new logos, datasets in existence tend to fail often, in the aspect of capturing the real-world logos, leading to an insufficient level of model performance.

Tackling these issues requires the combination of synthetic data generation with a typically trained ResNet50 deep learning model using a triplet loss function. Synthetic data can tend to expand training datasets for the improvement of model generalization, mainly for unseen logo recognition. The triplet loss function supports the learning potential of the model for distinct logo features and enhancement of similarity search accuracy. This method is ultimately aimed towards the improvement of logo recognition systems' accuracy, reliability, and scalability.

Chapter II

Literature Review

Logo recognition and similarity search have garnered significant attention in the computer vision community due to their wide-ranging applications in brand monitoring, intellectual property protection, and content-based image retrieval. Traditional approaches to logo recognition relied heavily on handcrafted feature extraction techniques, but recent advances in deep learning have substantially enhanced performance in terms of accuracy and robustness. This chapter reviews key research contributions in deep learning-based logo recognition, metric learning using triplet loss, synthetic data augmentation, and relevant real-world applications.

Early logo recognition methods relied on feature engineering techniques such as Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT), which were effective in controlled environments but lacked robustness under variations such as occlusion, rotation, and scaling (Dalal & Triggs, 2005; ?, ?). The emergence of convolutional neural networks (CNNs) transformed the field, with architectures like AlexNet (Krizhevsky, Sutskever, & Hinton, 2012) and VGGNet (Simonyan & Zisserman, 2014) demonstrating strong performance by automatically learning hierarchical features from data. However, deeper CNNs faced optimization challenges, notably the vanishing gradient problem, which limited their scalability.

To address this, He et al. introduced ResNet, which uses residual learning to enable the effective training of deeper networks (He, Zhang, Ren, & Sun, 2016). ResNet50, in particular, has become a widely used backbone for feature extraction due to its robustness and ability to preserve spatial information across layers. Several studies have shown that ResNet50 consistently outperforms other architectures on diverse logo datasets. While newer architectures such as Vision Transformers (ViTs) have gained popularity for image recognition tasks (Dosovitskiy et al., 2021), CNNs like ResNet50 continue to be the preferred choice for tasks that require high spatial fidelity, computational efficiency, and general robustness.

A major challenge in training deep learning models, including those for logo recognition, is the limited availability of large and diverse annotated datasets. This limitation affects the generalization ability of models when deployed in real-world scenarios. To mitigate this issue, researchers have employed various data augmentation techniques to artificially expand datasets. Traditional augmentation methods—such as flipping, scaling, and color jittering—have proven useful (Shorten & Khoshgoftaar, 2019), but recent innovations have explored generative approaches. AutoAugment, for example, learns augmentation policies directly from the data to optimize performance (Cubuk, Zoph, Mané, Vasudevan, & Le, 2019).

Generative Adversarial Networks (GANs) have revolutionized synthetic data generation, producing realistic images that can supplement limited datasets (Goodfellow et al., 2014). Studies have shown that GAN-generated data enhances CNN performance in scenarios with limited real samples (Radford, Metz, & Chintala, 2015). This is particularly useful in logo recognition, where gathering large volumes of annotated images is costly and time-consuming.

Similarity-based logo retrieval depends on learning compact and discriminative embeddings. Metric learning techniques, particularly those based on triplet loss, have shown promise in learning such embeddings. The FaceNet model introduced by Schroff et al. demonstrated how triplet loss can learn high-quality representations by minimizing the distance between similar samples while maximizing it between dissimilar ones (Schroff, Kalenichenko, & Philbin, 2015). This approach has been extended to various tasks, including image and signature retrieval, and has shown effectiveness in differentiating logos with subtle differences.

Gordo et al. applied triplet loss in large-scale image retrieval, achieving improvements in ranking precision and retrieval performance (Gordo, Almazán, Revaud, & Larlus, 2017). In logo recognition tasks, triplet loss enables the construction of meaningful embedding spaces where similar logos cluster together and dissimilar logos are separated, thereby improving both classification and retrieval accuracy.

Chapter III

Methodology

In this chapter, we present the methodological framework adopted to address the challenge of large-scale logo recognition and similarity retrieval. The goal is to learn an embedding space where logos that are visually and semantically similar are located close to each other, enabling scalable and efficient search.

To this end, we propose a pipeline that integrates synthetic data generation, multimodal feature extraction, dimensionality reduction, unsupervised clustering, and deep metric learning using a triplet-based architecture. The core idea behind this design is to avoid reliance on costly manual labeling by using pseudo-labels derived from unsupervised clustering.

3.1 Data Preparation

The data preparation stage forms the cornerstone of the proposed system for logo recognition and similarity retrieval. Its primary function is to convert raw multimodal data—namely synthetic logo images and their associated textual prompts—into structured, low-dimensional representations suitable for unsupervised clustering. This, in turn, facilitates the generation of weak labels for downstream deep metric learning.

Figure 3.1 illustrates the complete workflow of this stage.

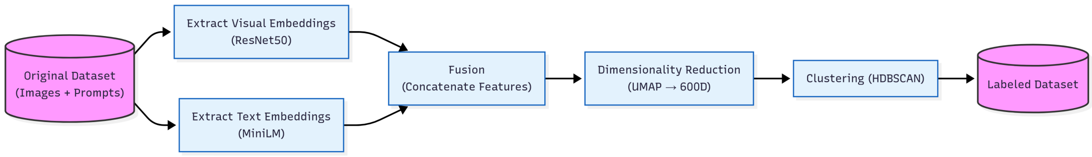

Figure 3.1: Overview of the data preparation pipeline: multimodal feature extraction, fusion, dimensionality reduction, and clustering

3.1.1 Multimodal Embedding Construction

The input to this stage is a dataset comprising 1,777,584 synthetic logo images, each annotated with a prompt describing its style, shape, or conceptual design. To jointly encode visual and textual information, we employ a dual-branch embedding approach.

Each image is processed using a frozen ResNet50 convolutional neural network pretrained on ImageNet. The final classification layer is removed, and the output from the last convolutional block is flattened to yield a 2048-dimensional visual embedding. In parallel, the textual prompt is embedded using the transformer-based MiniLM model (`all-MiniLM-L6-v2`), producing a 384-dimensional semantic vector.

The visual and textual embeddings are then concatenated into a unified feature vector:

$$\mathbf{z} = [\mathbf{v}; \mathbf{t}] \in \mathbb{R}^{2432}$$

This high-dimensional multimodal representation combines stylistic nuances from both image and language, thereby enabling a more expressive understanding of logos for subsequent analysis.

3.1.2 Dimensionality Reduction via UMAP

Although informative, the concatenated vectors are too high-dimensional for efficient clustering. To address this, we apply Uniform Manifold Approximation and Projection (UMAP) to project the embeddings into a 600-dimensional space:

$$\mathbf{z}' = \mathrm{UMAP}(\mathbf{z}) \in \mathbb{R}^{600}$$

UMAP is configured with cosine similarity as the distance metric and a minimum distance of zero to preserve fine-grained local structure. This step ensures that semantically similar logos are projected to nearby points in the latent space, which is crucial for reliable clustering.

3.1.3 Unsupervised Clustering with HDBSCAN

The reduced embeddings are subjected to unsupervised clustering using HDBSCAN, a density-based algorithm capable of discovering clusters with varying shapes and densities without requiring a predefined number of clusters. Initially, HDBSCAN identifies core dense regions and labels points in sparse areas as outliers. To ensure that every logo is assigned to a cluster, a secondary HDBSCAN pass is executed specifically on the outliers from the first pass. To avoid overlap, the resulting cluster IDs from this second pass are offset by the maximum cluster ID in the first pass.

The outcome of this two-stage clustering pipeline is a set of 209,310 distinct visual categories, each containing at least two logos. These pseudo-categories serve as surrogate labels that drive the sampling of training triplets in the metric learning phase.

3.1.4 Final Dataset Format and Stratified Partitioning

Upon completing the clustering process, the dataset is transformed from its original multimodal format—consisting of image-prompt pairs and their corresponding 2432-dimensional embeddings—into a more compact and task-oriented structure. Specifically, each data point is now represented by its raw image and an automatically assigned pseudo-category derived from the clustering phase:

$$\texttt{Dataset} = \{(\texttt{image}_i, \texttt{category}_i)\}_{i=1}^{N}$$

While this representation is considerably more lightweight and directly suited for metric learning, it introduces several significant challenges.

First, the sheer size of the dataset (1,777,584 images) poses a considerable computational burden for both memory management and I/O efficiency. Furthermore, the clustering process results in an extremely large number of distinct categories—over 209,000 in total. This high granularity, although beneficial for capturing nuanced variations among logos, introduces a substantial level of **class imbalance**, with many categories containing only a handful of samples. Such sparsity severely complicates the formation of meaningful training triplets and may lead to degenerate cases where positive or negative examples cannot be reliably sampled.

To ensure that the resulting dataset is both semantically structured and suitable for training, we perform a stratified partitioning strategy guided by cluster integrity. Categories that contain fewer than two images are discarded, as they cannot contribute to valid anchor–positive pairs. For all remaining categories, the associated samples are randomly shuffled and then divided into training, validation, and testing sets. Approximately 70% of the samples are allocated to training, up to 10% to validation, and the remaining 20% to testing—subject to the constraint that each split must include a minimum of two images per category to maintain triplet validity.

The final output of this process is a refined and operational dataset comprising 1,279,860 training samples, 142,207 validation samples, and 355,517 test samples. While the pseudo-categories serve as weak labels, their derivation through unsupervised clustering allows the system to scale without manual annotation. However, the inherent imbalance and long-tail distribution of categories remain a central challenge in ensuring fair learning and generalization—issues that are directly addressed through robust sampling strategies and deep metric learning in the subsequent stages.

3.2 Triplet Network Architecture

The core of the proposed similarity retrieval system is a deep metric learning framework based on a Triplet Network. The objective of this architecture is to learn an embedding space where visually and semantically similar logos are mapped close together, while dissimilar ones are projected farther apart. The triplet-based approach enables the model to learn fine-grained similarity relationships without relying on explicit class labels.

Figure 3.2 illustrates the architecture of the Triplet Network used in this study.

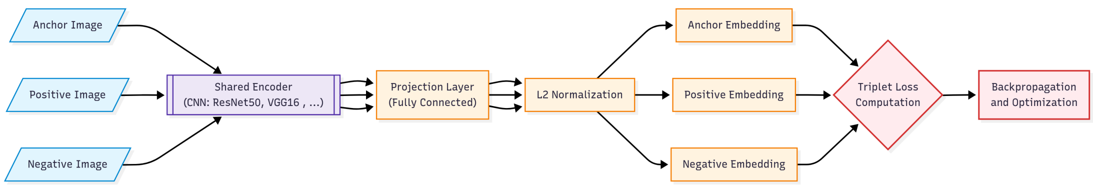

Figure 3.2: Triplet Network architecture for metric learning. Shared weights are applied across all branches to ensure consistent representation learning.

The network takes as input three images forming a triplet: an anchor, a positive (from the same pseudo-category as the anchor), and a negative (from a different pseudo-category). Each image is passed through the same convolutional backbone—either ResNet50, VGG16, or EfficientNet-B0—ensuring that all branches share identical weights. This design enforces consistent feature extraction across samples and prevents the model from overfitting to specific image positions within the triplet.

The output from the CNN encoder is a high-dimensional feature vector that captures the essential visual characteristics of the logo. This vector is then passed through a projection head composed of fully connected layers, batch normalization, ReLU activation, and dropout. The final layer of the projection head is a dense layer whose output is normalized to lie on the unit hypersphere using L2 normalization. This normalization facilitates the use of cosine distance in the embedding space.

Let $f(\cdot)$ denote the full encoding function (CNN + projection + normalization), and let a, p, and n be the anchor, positive, and negative samples, respectively. The triplet network aims to satisfy the constraint:

$$\|f(a) - f(p)\|_2^2 + \alpha < \|f(a) - f(n)\|_2^2$$

where α is a fixed margin (set to 1.0). The associated loss function is the Triplet Margin Loss:

$$\mathcal{L}_{\text{triplet}} = \max\left(0, \|f(a) - f(p)\|_2^2 - \|f(a) - f(n)\|_2^2 + \alpha\right)$$

This loss encourages the network to minimize the distance between the anchor and the positive sample while ensuring that the negative is pushed at least α units away.

The projection head plays a critical role in learning a task-specific embedding space that is compact, discriminative, and suitable for nearest neighbor retrieval. It also serves as a bottleneck that controls the final dimensionality of the embedding vector (e.g., 256D), which has practical advantages in terms of storage and inference speed.

Overall, the Triplet Network transforms raw image inputs into fixed-length, semantically meaningful vectors that can be directly used in similarity search applications via simple distance-based retrieval.

3.3 Training Procedure

The training procedure aims to optimize the Triplet Network to learn a discriminative and semantically meaningful embedding space. In this space, visually or conceptually similar logos are mapped to proximate vectors, while dissimilar logos are pushed farther apart. Since manual labeling is not available, we rely on the pseudo-categories generated through unsupervised clustering to serve as weak supervisory signals for constructing training triplets.

Each training sample is composed of a triplet: an anchor image a, a positive image p belonging to the same pseudo-category, and a negative image n from a different category. These triplets are sampled dynamically during training, ensuring diversity and reducing the risk of overfitting. A custom PyTorch-based dataset class is used to generate and collate triplets on-the-fly based on the precomputed category indices.

3.3.1 Projection Head Configuration

Following the frozen convolutional backbone (ResNet50, VGG16, or EfficientNet-B0), the model includes a trainable projection head designed to transform the extracted visual features into a lower-dimensional space suitable for metric learning. The projection head consists of two fully connected blocks with batch normalization, ReLU activation, and dropout layers.

In our implementation, the architecture is defined by the following unit sizes:

$$\texttt{ff\_block\_units} = [512, 256]$$

The first linear layer maps the backbone output (e.g., 2048 for ResNet50) to 512 dimensions. The second layer projects this intermediate representation to the final embedding size of 256 dimensions. A dropout rate of 0.3 is applied between layers to improve generalization. The output is finally normalized using L2 normalization to lie on the unit hypersphere, facilitating cosine-based similarity comparisons during retrieval.

This design balances capacity and compression, allowing the model to retain essential semantic details while generating compact, retrieval-efficient vectors.

3.3.2 Optimization and Hyperparameters

Training is conducted using the Adam optimizer with an initial learning rate of 1×10^{-4} and a weight decay coefficient of 0.01. The model is trained for 10 epochs with a batch size of 1024 triplets. The backbone is frozen throughout training to preserve its pretrained features and reduce computational overhead. The training process is monitored via TensorBoard, and the model checkpoint with the lowest validation loss is saved for evaluation and inference.

Table 3.1 summarizes the key hyperparameters and configurations used.

Table 3.1: **Summary of training hyperparameters and model settings**

Parameter	**Value**
Optimizer	Adam
Learning Rate	1×10^{-4}
Weight Decay	0.01
Epochs	10
Batch Size	1024
Loss Function	Triplet Margin Loss
Triplet Margin (α)	1.0
Backbone Network	ResNet50 / VGG16 / EfficientNet-B0
Backbone Status	Frozen
Projection Head Structure	[512, 256]
Embedding Size	256
Dropout Rate	0.3
Evaluation Strategy	Per Epoch
Logging	TensorBoard

3.3.3 Training Stability and Loss Curves

To monitor the convergence behavior and generalization capacity of each backbone, we tracked both training and validation loss throughout all epochs. Figures 3.3, 3.4, and 3.5 display the corresponding loss curves for ResNet50, VGG16, and EfficientNet-B0 respectively.

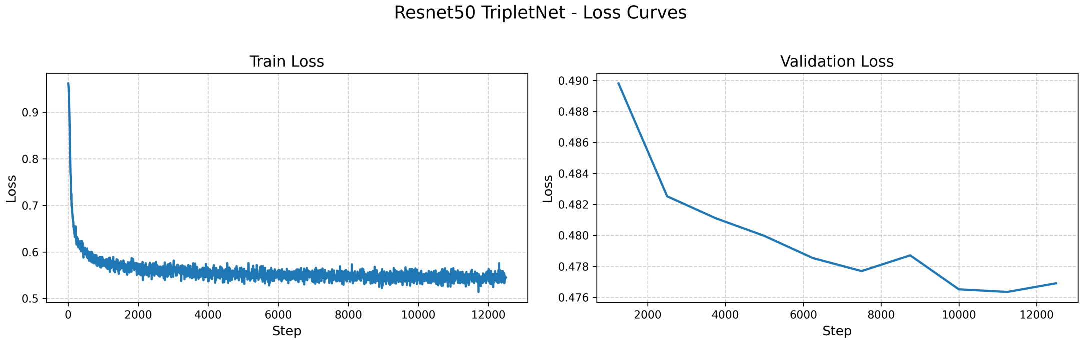

Figure 3.3: ResNet50: Training and validation loss across epochs

The ResNet50 backbone shows stable and rapid convergence, with a clear reduction in validation loss over time. The consistent gap between training and validation losses suggests strong generalization and low overfitting.

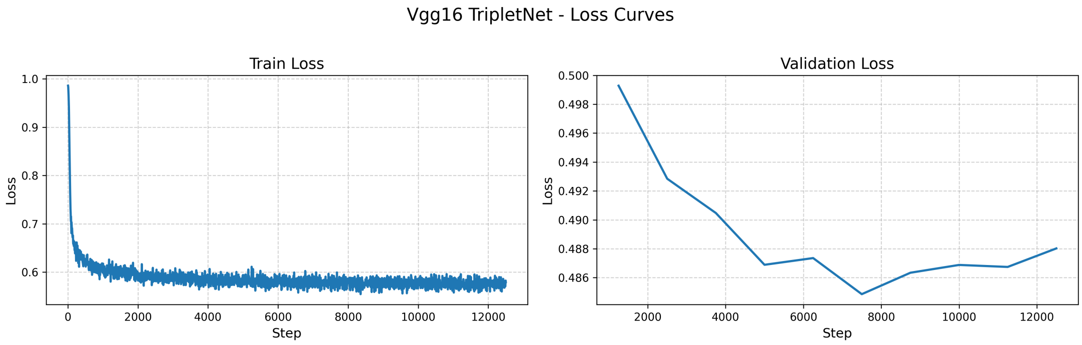

Figure 3.4: VGG16: Training and validation loss across epochs

VGG16 demonstrates a slower but stable decline in loss. Its final validation loss remains slightly higher than that of ResNet50, indicating weaker regularization or representational power.

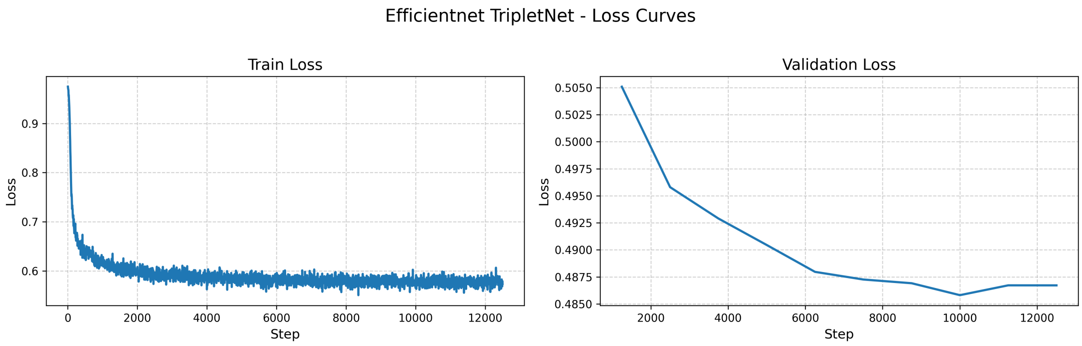

Figure 3.5: EfficientNet-B0: Training and validation loss across epochs

EfficientNet-B0 begins with higher initial loss and shows consistent improvement but with occasional fluctuations. This suggests sensitivity to sample variance and a possible need for longer training or learning rate adjustment.

These visualizations confirm that all backbones are capable of learning from the pseudo-labeled data, but that ResNet50 exhibits superior training stability and convergence properties.

3.3.4 Training Challenges

Training a metric learning model over a dataset of this scale introduces several non-trivial challenges. The extreme class imbalance—due to the long-tail distribution of the 209,310 pseudo-categories—makes it difficult to sample uniformly meaningful triplets. Many categories contain only a small number of samples, limiting the availability of positive examples and increasing the risk of degenerate triplets.

In addition, the sheer volume of data requires efficient memory management and data loading techniques. Triplet generation must be both dynamic and scalable, which is achieved using custom collators and category-index caching strategies.

Despite these difficulties, the training procedure remains effective due to: (1) the high semantic resolution provided by the projection head; (2) robust sampling mechanisms; and (3) extensive use of parallelization on GPUs. The resulting model learns a generalizable embedding space that forms the basis for high-performance similarity retrieval in the following stages.

3.4 Embedding Retrieval Pipeline

After the Triplet Network has been trained and validated, the final step in the pipeline is the deployment of the model for inference and similarity retrieval. In this stage, the trained network is used to encode unseen logo images into fixed-length embedding vectors. These embeddings are then used to perform fast and accurate nearest-neighbor retrieval over a precomputed index of known logos.

The overall retrieval pipeline is illustrated in Figure 3.6.

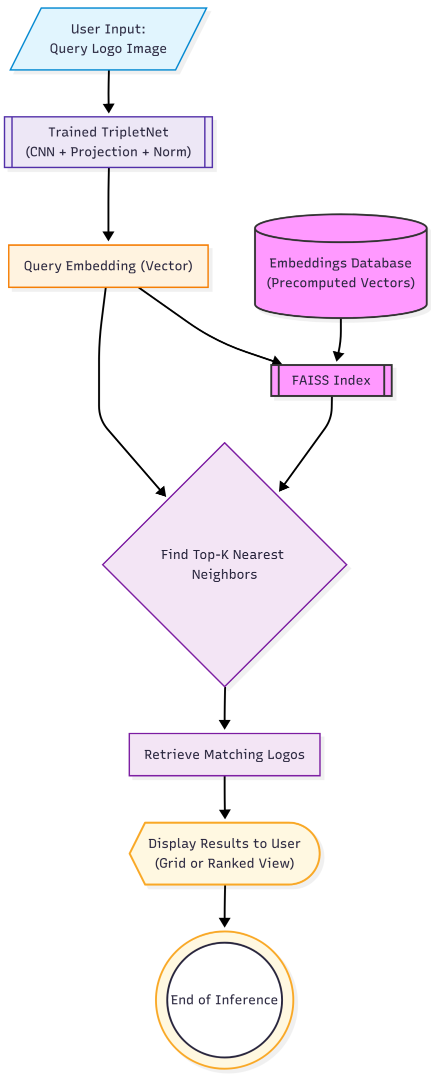

Figure 3.6: Inference-time retrieval pipeline using the trained Triplet Network and FAISS-based similarity search.

Given a query image, the system performs the following steps:

1. The image is passed through the trained Triplet Network to obtain a normalized embedding vector in the 256-dimensional latent space.
2. The resulting embedding is compared against a FAISS index containing embeddings of training images.
3. A similarity score is computed using cosine distance or L2 norm.
4. The top-K most similar logos are retrieved based on the similarity ranking.

This architecture is designed to be highly scalable and efficient. Since all embeddings are normalized, the retrieval process can rely on cosine similarity, which is computationally efficient and effective for high-dimensional vector spaces.

FAISS (Facebook AI Similarity Search) is used as the core indexing engine due to its ability to handle millions of high-dimensional vectors while offering sublinear query time. The embeddings are indexed offline and loaded into memory during runtime for real-time inference.

The quality of the retrieval depends entirely on the structure of the learned embedding space. If the training procedure successfully places semantically similar logos close together, then the retrieval results will reflect true visual or conceptual similarity. As such, the embedding retrieval pipeline directly leverages the discriminative power of the learned representations to support use cases such as logo search, duplicate detection, and content moderation.

Chapter IV

Results and Discussion

To assess the effectiveness of the trained Triplet Network models in capturing logo similarity, we adopt an evaluation strategy based on vector-space retrieval. Instead of performing classification or segmentation, the system is evaluated by its ability to retrieve semantically or visually similar logos from a database, given a query image.

At inference time, each test logo image is passed through the trained Triplet Network to generate a normalized 256-dimensional embedding vector. This embedding is then compared to a precomputed FAISS index of embeddings obtained from the training set. The goal is to retrieve the top-K most similar logos based on cosine similarity or L2 distance.

The ground-truth relevance of retrieved logos is determined using the pseudo-categories generated during the unsupervised clustering phase. If a retrieved image shares the same pseudo-category as the query image, it is considered relevant.

To quantify retrieval performance, we compute the following standard ranking-based metrics:

- **Precision@K:** the proportion of top-K retrieved logos that are relevant.

- **Recall@K:** the proportion of relevant logos that appear in the top-K results.

- **F1@K:** the harmonic mean of Precision@K and Recall@K, offering a balanced view of performance.

These metrics are computed at multiple cutoff levels (K = 1, 5, 10) to evaluate both narrow and broader retrieval quality. The use of FAISS indexing allows scalable evaluation over hundreds of thousands of embeddings, ensuring that results are both practical and statistically robust.

The subsequent sections present the quantitative results of this evaluation and provide detailed insights into model behavior and architectural trade-offs.

4.1 Quantitative Results

The evaluation was conducted on the full test set consisting of 355,517 logo samples. For each test image, the top-K nearest neighbors were retrieved from the training set using the FAISS index. The retrieved results were then compared to the pseudo-category label of the query image to compute Precision@K, Recall@K, and F1@K at $K = 1, 5, 10$.

Table 4.1 presents the quantitative performance metrics for each of the three backbone architectures used in the Triplet Network: ResNet50, VGG16, and EfficientNet-B0.

Table 4.1: Retrieval performance on test set using FAISS-based similarity search

Model	Top-K	Precision@K	Recall@K	F1@K
ResNet50	1	0.1827	0.0462	0.0737
ResNet50	5	0.1844	0.2344	0.2064
ResNet50	10	0.1872	0.4847	0.2701
VGG16	1	0.1810	0.0452	0.0723
VGG16	5	0.1812	0.2266	0.2014
VGG16	10	0.1832	0.4641	0.2627
EfficientNet-B0	1	0.1810	0.0453	0.0724
EfficientNet-B0	5	0.1811	0.2267	0.2013
EfficientNet-B0	10	0.1829	0.4630	0.2622

From the results, we observe that all three architectures perform comparably, especially at lower values of K. However, the ResNet50 backbone consistently achieves slightly higher scores across all metrics and cutoff levels. At $K = 10$, for instance, ResNet50 achieves an F1 score of 0.2701, compared to 0.2627 for VGG16 and 0.2622 for EfficientNet-B0.

These differences, while modest, suggest that ResNet50 may produce more semantically coherent embeddings, particularly for broader retrieval scopes. Notably, recall increases substantially as K increases, indicating that many relevant results are indeed captured but may be ranked lower in some models.

Precision remains relatively stable across models, suggesting consistent relevance among the top-ranked matches. Overall, the quantitative results affirm that all architectures are capable of learning meaningful similarity relationships, with ResNet50 demonstrating a slight advantage.

4.2 Qualitative Interpretation

While quantitative metrics provide a statistical overview of retrieval performance, a qualitative examination reveals important insights into the practical behavior of the models and their ability to learn meaningful semantic structures.

One of the most noticeable trends is the sensitivity of retrieval performance to the density and size of the pseudo-categories. Categories with a larger number of representative samples tend to yield higher Precision@K and Recall@K scores. This is because such categories allow for more diverse positive examples during training, which helps the model generalize better to unseen logos within the same conceptual group.

Conversely, the performance degrades for long-tail categories that contain very few samples—often only two or three logos. In these cases, the network struggles to form robust intra-class representations, which in turn reduces the likelihood of retrieving semantically aligned logos during inference.

Another important observation concerns logos with similar global shapes but differing fine-grained stylistic details. The models occasionally fail to distinguish between such logos, especially when the differences are subtle (e.g., thin lines, color tone, font variation). This points to the limitation of using purely visual embeddings without more fine-tuned local attention mechanisms. Nevertheless, the inclusion of text prompts during the clustering stage (via multimodal embeddings) appears to have helped anchor the embeddings to more semantically structured regions of the latent space.

Among the architectures, ResNet50 tends to produce slightly more compact clusters in the embedding space, leading to better semantic separation. EfficientNet-B0, while computationally more efficient, sometimes suffers from embedding collapse in rare classes—where all samples are projected too closely together, making distinction difficult.

Finally, it is worth noting that the model is particularly effective in identifying duplicates and near-duplicates, even when logos undergo transformations such as scaling, rotation, or minimal re-coloring. This suggests that the learned embedding space captures not only identity but also structural invariance, which is essential for reliable logo recognition and similarity search.

Chapter V

Conclusion and Future Perspective

5.1 Conclusion

This research presents a scalable and efficient pipeline for logo recognition and similarity search, leveraging synthetic data, multimodal embeddings, unsupervised clustering, and deep metric learning. By utilizing a dataset of over 1.7 million synthetically generated logos paired with descriptive prompts, we overcome the limitations of conventional real-world logo datasets, such as limited variety, annotation cost, and poor generalization.

A key innovation of this work lies in the use of multimodal embeddings—combining visual features from ResNet50 with semantic features from MiniLM—to construct a rich 2432-dimensional representation of each logo. This embedding is then reduced and clustered to generate over 209,000 pseudo-categories, which serve as weak supervision for triplet-based training. The resulting Triplet Network learns a discriminative embedding space where similar logos are embedded nearby, enabling fast and accurate retrieval using FAISS indexing.

Quantitative results demonstrate that the proposed method achieves reliable retrieval performance, with ResNet50 outperforming VGG16 and EfficientNet in most scenarios. Qualitative analysis further confirms the model's ability to capture semantic similarity and visual coherence across diverse logo designs.

Overall, the proposed system demonstrates the feasibility and effectiveness of combining synthetic data generation with weakly supervised metric learning for scalable visual similarity retrieval tasks.

5.2 Limitations

Despite its strengths, the system faces several limitations. First, the reliance on unsupervised clustering introduces noise into the pseudo-labels, which may affect training stability, particularly in rare or visually ambiguous categories. Second, the long-tail distribution of pseudo-categories results in imbalanced training data, limiting the diversity of positive pairs in small clusters. Third, the frozen CNN backbones constrain the capacity to adapt to subtle domain-specific features unless fine-tuned.

Furthermore, while multimodal embeddings were used during the clustering phase, the final retrieval system operates solely on visual inputs. Incorporating multimodal representations at inference could further improve semantic alignment.

5.3 Future Perspective

Future work can expand on this foundation in several directions. One promising direction is the use of self-supervised or contrastive pretraining to reduce reliance on synthetic prompts and improve generalization. Exploring transformer-based vision architectures (e.g., Vision Transformers) may yield richer representations, particularly for logos with complex stylistic cues.

Additionally, integrating multilingual or generative prompt modeling may improve clustering accuracy by providing higher-quality semantic anchors. Extending the system to support cross-modal retrieval—retrieving logos based on text descriptions or sketches—would further broaden its applicability in commercial and creative domains.

Finally, deployment-focused improvements such as lightweight model distillation, quantization, and real-time retrieval interfaces will be essential for enabling industry-scale adoption.

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